

Deepak Mallampati

APPLIED AI / FORWARD DEPLOYED ENGINEER

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US-based (Kent, OH) -- F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI engineer with **4+ years** deploying and configuring production AI agents and LLM systems into real team workflows — from integration to measurable adoption. Built and integrated **25+ AI agents** and **12 APIs/microservices** embedded across **20+ internal teams**, serving **100,000+ requests daily**. Architected agentic conductor-subagent systems (**42%** token reduction), production RAG pipelines (**40%** hallucination reduction, **92%** precision), and repeatable LLM evaluation frameworks (RAGAS, BERTScore) that turn fuzzy quality into measurable success criteria. Comfortable mapping a team's workflow, customizing agents to it through prompts and integrations, and writing the playbook that makes adoption stick.

CORE COMPETENCIES

AI Agent Deployment & Configuration . Forward-Deployed Delivery . Prompt Engineering & Customization . SaaS / API Integrations & Data Pipelines . Multi-Agent & Agentic Orchestration . Production RAG . Repeatable Implementation Patterns & Success Criteria . Stakeholder Enablement & Adoption . Python / SQL / FastAPI . LLM Evaluation & Observability

EXPERIENCE

Progress Solutions

Jul 2025 -- Present

AI Engineer -- USA

- Architected production **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing complex objectives into context-scoped subagents that execute autonomously — reducing token consumption **42%** while sustaining task accuracy across multi-step workflows.
- Engineered high-performance **RAG** pipelines combining dense vector retrieval with cross-encoder re-ranking to lift answer relevance and materially reduce hallucination on domain-specific queries.
- Established a structured **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom domain metrics) with latency/accuracy dashboards — the repeatable success criteria that benchmark iterations and surface regressions before release.
- Built fault-tolerant **automation infrastructure** with scheduled orchestration and exponential-backoff retry logic, reducing failed production runs **60%**.
- Optimized inference cost and latency through prompt compression, semantic caching, and model routing while holding SLA targets.

Vanguard

Jan 2024 -- Jan 2025

AI Engineer Intern -- USA

- Developed and integrated **AI agents and backend services** for Vanguard's internal AI platform across **25+ AI agents** and workflows, wiring them into production data pipelines and observability tooling.
- Optimized LLM prompts and evaluated **GPT-4o, Claude Sonnet, and Gemini**, improving task success rates **27%** and informing platform model-selection decisions.
- Built **12 APIs and microservices** powering new AI capabilities, cutting p95 latency from 1.5s to **800ms** and accelerating feature integration **40%**.
- Partnered with product, platform, and data engineering teams to deliver scalable LLM applications supporting **100,000+ requests daily** across **20+ internal teams** — embedding with each team to translate their needs into shipped AI capabilities.
- Engineered content controls and safeguards for policy-compliant responses, reducing unsafe outputs **42%**.

Emerson

Jun 2022 -- Apr 2023

ML Trainee -- India

- Built supervised regression/classification models for equipment-failure prediction, anomaly detection, and capacity forecasting on operational pipeline-storage data — embedding with SMEs to turn the process into a model-ready spec.
- Fine-tuned **BERT and RoBERTa** for entity extraction and text classification on domain-specific datasets, achieving **89% F1** on custom NER tasks.

ML Intern -- India

- Designed an end-to-end video summarization and highlight pipeline (speech-to-text, transformer summarization, timestamp-aligned clip extraction), deployed across **5,000+ videos** and reducing manual editing effort **70%**.

PROJECTS

career-ops autonomous agent pipeline

An AI-driven job-search pipeline that scans listings, scores fit, and generates tailored applications overnight without supervision — spotting a repetitive workflow and shipping an end-to-end automated tool. Direct analog to the deploy-an-agent-into-a-workflow loop.

Python, LLM agents, workflow orchestration, scheduled automation

MangaLens shipped -- Chrome Web Store

A shipped Chrome extension delivering real-time AI translation of manga and webtoons across **29+ sites** — turning a multi-step manual chore into a single inline action, with real users in production.

TypeScript, AI vision providers, multi-provider abstraction

Privacy-Preserving Clinical ML Pipeline

A framework combining k-anonymity, l-diversity, and differential privacy that cut record-linkage re-identification risk from **3.38% to 0.32%** while retaining **99%** of baseline predictive performance.

Python, scikit-learn, differential privacy, privacy-utility benchmarking

EDUCATION

M.S. in Computer Science -- Kent State University, Ohio

2025 | GPA 3.70 / 4.0

TECHNICAL SKILLS

AI Agents & LLMs: GPT-4o, GPT-5, Claude Sonnet, Gemini, LLaMA, LangChain/LangGraph, multi-agent systems, prompt engineering, structured outputs, RAG architecture, fine-tuning (LoRA, QLoRA, PEFT)

Integrations & Backend: Python, SQL, JavaScript/TypeScript, FastAPI, REST APIs, microservices, Kafka, Spark, PostgreSQL, MongoDB, Redis

Evaluation & MLOps: RAGAS, BERTScore, A/B testing, drift detection, MLflow, Weights & Biases, model monitoring & observability, Docker, CI/CD

Cloud: AWS (Bedrock, SageMaker, Lambda, EC2, S3), Azure OpenAI, GCP Vertex AI

Security & Compliance: HIPAA, SOC 2, PCI DSS, OAuth 2.0/JWT/RBAC, PII masking, prompt-injection mitigation, audit logging

Work Authorization: F-1 OPT -- authorized to work in the United States now; future sponsorship required